Validation Report

Test against a test mailbox with known hashes

Environment

Date (UTC / local)	2026-11-08 18:35:41 UTC / 2026-11-08 19:35:41 CET
Duration	5 s
Tool	FED - Forensic Email Duplicator 1.0.0.dev0
SHA-256 of program	`967692e8b03b93b27ace0d10078d4507018d39511c18c255652b7627a3fa72b8`
Source revision (Git)	`d1d8447eb45d837c1901880639ff09f32c32c83c`
Python	3.13.18
Operating system	Linux-6.18.44-fc-v310-x86_64-with-glibc2.39
IMAP server	Dovecot 2.3.21 (47349e2482)
OpenSSL	OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)

Result

All checks performed passed.

PASSED: 15 FAILED: 0 NOT TESTED: 0

Checks

V01 Read-only access — PASSED

Expected: Mailbox on the server unchanged after acquisition: all Maildir files (content, name incl. flags, time) identical, STATUS before = after in every folder.

· 14 Maildir files compared; 0 folders with STATUS difference

V02 Blocking of writing IMAP commands — PASSED

Expected: STORE, COPY, MOVE, EXPUNGE, APPEND, DELETE, RENAME, CREATE, SELECT and FETCH without PEEK (would set the seen flag) are refused before being sent.

· STORE: refused
· COPY: refused
· MOVE: refused
· EXPUNGE: refused
· APPEND: refused
· DELETE: refused
· RENAME: refused
· CREATE: refused
· SELECT: refused
· FETCH BODY[]: refused

V03 Completeness — PASSED

Expected: All folders (including empty ones) found; number of messages per folder as expected.

· Archiv: 1 / 1
· Entwürfe: 1 / 1
· Gesendet: 1 / 1
· Groß: 1 / 1
· INBOX: 5 / 5
· INBOX/Rechnungen: 1 / 1
· Leer: 0 / 0
· Projekte: 0 / 0
· Projekte/2025: 1 / 1
· Projekte/2026: 1 / 1
· Verband: 2 / 2

V04 Bit-exact acquisition — PASSED

Expected: SHA-256, MD5 and size of every .eml match expected_hashes.csv (incl. 8-bit Latin-1, no Date header, 4 MB message fetched in chunks).
· 14 of 14 messages bit-exact

V05 Receipt time as file time — PASSED

Expected: File time and file name of every .eml correspond to the receipt time (INTERNALDATE).
· 14 files checked

V06 Cancel and resume — PASSED

Expected: A cancelled attempt leaves no partial files; resuming loads only what is missing, final result bit-exact.
· Attempt 1: cancelled, 5 saved; partial files afterwards: 0
· Attempt 2: done, 9 new, 5 existing and re-hashed

V07 Detection of modified working files — PASSED

Expected: A .eml modified afterwards is detected on resume, logged and downloaded again.
· modified file: test@example.de/Verband/20261105T090005Z_UID00000001.eml
· logged: 1× existing_file_invalid
· downloaded again: 1, bit-exact afterwards: yes

V08 ZIP archive — PASSED

Expected: ZIP verified; content bit-exact; folder structure incl. umlaut and empty folders; .sha256 file correct.
· VAL-1_20261108T183542Z.zip: 14 .eml, verified: yes
· .sha256 file = file hash: yes
· content bit-exact: yes
· folders in ZIP: 11; checked: test@example.de/Entwürfe/, test@example.de/Leer/

V09 Detection of modified ZIP content — PASSED

Expected: A message modified inside the ZIP is detected by verification.
· Hash abweichend: test@example.de/Archiv/20251218T133002Z_UID00000001.eml

V10 Tamper protection of the log — PASSED

Expected: Log append-only (UPDATE/DELETE refused); hash chain intact; modifying an entry is detected.
· hash chain of the acquisition: intact (20 entries)
· UPDATE: refused
· DELETE: refused
· copy with modified entry #2: detected (at #2)

V11 RFC 3161 timestamp — PASSED

Expected: Timestamp for the ZIP hash received; hash and nonce match; signature valid with openssl ts -verify.
· time per service: 2026-11-08T18:35:42Z, serial 2, accuracy 1 s
· timestamped hash = ZIP SHA-256: yes
· openssl ts -verify: Verification: OK
· Note: test timestamp service with its own test CA (this checks FED, not a public service)

V12 Reports and output hashes — PASSED

Expected: report.txt/report.pdf and log.csv created; all output hashes recorded in the case file match the files.
· 5 output files checked against the case file

V13 Umlauts in address, domain and password — PASSED

Expected: Account bärbel.böhmer@ärzte.de (IDN domain, umlauts in name and password) signs in and is acquired bit-exactly; the ZIP folder carries the address in Unicode (UTF-8 flag).
· domain for DNS/TLS: xn--rzte-koa.de
· login and acquisition: done, 14 messages
· folder in ZIP: bärbel.böhmer@ärzte.de; UTF-8 flag: yes
· bit-exact: yes

V14 Authenticity check (DKIM) — PASSED

Expected: Validly signed message = valid, message modified after signing = invalid, others = not signed; the public key is preserved with SHA-256 in the case file; the provider's Authentication-Results are evaluated.

· results: {'pass': 1, 'none': 12, 'fail': 1}
· validly signed: pass; modified: fail (Text verändert (Body-Hash stimmt nicht))
· key preserved: yes (fed2026._domainkey.dkim.example)
· Authentication-Results evaluated: yes

V15 Targeted partial acquisition (content filter) — PASSED

Expected: With subject filter “Rechnung, übermorgen” (incl. umlaut) exactly the two matching messages are acquired; with an additional sender filter (and) none; criteria logged and shown in the report as partial acquisition.

· acquired: ['AW: Termin Donnerstag übermorgen', 'Rechnung 2026-1187']
· sender AND subject: 0 messages
· logged: 1×, in report: yes

Test mailbox

Folder	Received (UTC)	Size	SHA-256	Feature
INBOX	2026-11-05T07:12:44Z	556	947894308840a0b78704521bddbfaa2451a148d7…	Anlage PDF, gelesen
INBOX	2026-11-04T15:03:09Z	277	26125c8fc8a55f97415d90c51c04ab049dfd7b3c…	Umlaut im Betreff, 2 CC
INBOX	2026-11-03T09:00:00Z	207	29dd33790ee1349913a426dce808db0e9f322974…	8bit Latin-1, Bytes muessen unveraendert bleiben
INBOX	2026-11-02T12:30:00Z	130	25e1214dcd1321d9f3dbb3110eef0ba7c0480212…	ohne Date-Header
INBOX	2026-11-01T18:00:00Z	604	062bafe547e688ca6d85d71df1cf1aa221f63e62…	1 Anlage + 1 inline
INBOX/Rechnungen	2026-10-30T08:47:21Z	158	994c156f4a1b29345528b3ff02876ec02f361e70…	
Projekte/2025	2025-12-01T10:00:00Z	114	3ba5be665a75a52fea81745754cdbaa53f936e56…	
Projekte/2026	2026-02-01T10:00:00Z	114	5824abefa173dc7d4616b63c631d568e3e80417c…	
Entwürfe	2026-11-06T09:00:00Z	90	7785922a33dd51c893b737733fddca398f922d79…	Umlaut im Ordner
Gesendet	2026-10-21T10:02:17Z	176	3bc026d800ea14769a3b814b34566c4da2bed705…	nur BCC
Archiv	2025-12-18T13:30:02Z	176	6ccede1fc4fd1b169158c361ef4cb9684f1d239a…	per BCC erhalten
Groß	2026-09-01T08:00:00Z	4305058	0e4d197804045e079f72e28676c3b967aa5a8e70…	ca. 4 MB, stueckweises Laden
Verband	2026-11-05T09:00:05Z	988	044adfb7fd6338524bcd344be5596afd1e9036c5…	DKIM gueltig signiert, Authentication-Results
Verband	2026-11-06T09:00:05Z	806	a720fb1ce8785dd900ef3baf4f8ca0e363b21848…	DKIM-Signatur, Text nachtraeglich veraendert